

Nikil PS

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Summary

Software Engineer with 2+ years of experience delivering production-grade backend, AI, and full-stack solutions. Proven expertise in Python, FastAPI, React/TypeScript, WebSocket, Docker, and LLM integration (LangChain, Ollama). Strong background in computer vision, NLP, and embedded systems, with a Master's in Artificial Intelligence.

Skills and Certificates

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Frameworks/Libraries: FastAPI, React, Vite, python-chess, Stockfish, TensorFlow, PyTorch, OpenCV, Scikit-learn, Pandas, Keras, LangChain, ChromaDB, Next.js

Domains: Full-stack Development, AI/ML, NLP, Computer Vision, Embedded Systems, Robotics

Tools: Docker, Kubernetes, Linux, AWS, SVN, WebSocket, Vite, Postman, Git, Ollama

Certificates: NLP Specialization, CAN and CAN FD, Embedded C, Generative AI Specialisation

Experience

Software Engineer, Multimatic Electronic System – Cambridge,GB 2023 - Present

- Designed and containerized a Python-based key generation service using Docker and Kubernetes CI/CD, improving ECU security compliance for 1,200+ vehicles.
- Automated software baseline and compliance auditing workflows (Python + SVN APIs) for ASPICE Level 2 certification, achieving 100% compliance across 3 projects.
- Developed a full-stack automation platform (Node.js, Python, PostgreSQL, Svelte) using SVN and JIRA API to streamline commit reviews and traceability, improving engineering efficiency by 67%.
- Designed and deployed multiple Copilot-based automation agents for process monitoring, achieving 80% efficiency gains across the development lifecycle.
- Developed and deployed embedded C software for Infineon TC364x ECUs, ensuring ISO26262 and MISRA C compliance for production vehicles.
- Refactored and validated RNG algorithms in C to meet NIST SP 800-22 standards, ensuring cryptographic security for 800+ ECUs.
- Implemented real-time WebSocket-synchronized features and LLM-powered analytics in internal tools, enhancing user experience and data insights.

Education

University of Manchester, BS(Hons) Mechatronics

June 2021

- First Class: 77%

University of Edinburgh, MSc Artificial Intelligence

Nov 2022

- MSc Dissertation: 83%

Projects

Chess AI Multiplayer App

- Built a local, full-stack chess application with React18 + TypeScript, Vite, and FastAPI, integrating Stockfish16+ via UCI and python-chess for AI play.
- Delivered real-time, WebSocket-synchronized gameplay, custom AI difficulty, move-rating feedback, and comprehensive post-game analysis.
- Supported local multiplayer over LAN and hot-seat modes, with Docker-Compose for reproducible deployment.

Wingman: Local LLM RAG Agent

- Developed a local Retrieval-Augmented Generation (RAG) system using Next.js, Python, Ollama, LangChain, Docker, and ChromaDB for secure, offline document search.
- Designed modular backend APIs enabling real-time vector search and local LLM context retrieval, supporting multi-format document ingestion.

MSc Dissertation: Automated Recycling Sorting

University of Edinburgh

- Created a real-time waste classification system with VarifocalNet on the MMDetection API, achieving high-speed inference and 5-class object detection.
- Improved mAP by 244%, reduced training annotation by 70%, and enhanced generalizability across object classes.

BEng Dissertation: Driver Action Prediction

University of Manchester

- Built an SVM classifier to predict driver actions 500ms ahead using kinematic data, achieving a 90% F1-score in simulated real-time scenarios.
- Integrated real-time elbow kinematics from motion capture to enhance prediction robustness and reduce misclassification of spontaneous movements.